

Socio-Cultural Review of Cancer Across Different Continents

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Abstract

Cancer perception is different across each region of the world, which can lead to late diagnosis and death, especially in cancers associated with greater stigma. These perceptions are largely affected by sociocultural, economic, and healthcare infrastructure of the region. It is crucial that effective methods are implemented, so that patients do not avoid getting diagnosed and treated. Misconceptions, stigmas, myths and taboos lead the patients to have fear, shame and self distrust. Some patients may mistrust the medical facilities due to previous trauma. Cancer can cause social dilemmas, which may trigger mental health problems. Different regions have different social/cultural issues, which impact patients negatively. This study looks into each region in a detailed manner and provides personalized solutions for each area.

Introduction

Cancer Overview

Cancer is the second leading cause of death in the United States and globally (“The Global Cancer Burden,” n.d.). In fact, in 2022, it was predicted that around 609,360 people will die from cancer in the United states. Cancer is a complex and devastating condition marked by uncontrolled cell proliferation and quick spread throughout the body, a process called metastasis. The aberrant growth and division of cells leads to malignant tumors which invade and infiltrate neighboring tissue and organs. Systemic failure occurs when the organs no longer function properly. When cancer is in an early stage, it can be curable, however once the cancer has metastasized, the chances of recovery diminish significantly even with today’s technology. Therefore, it is important to spread awareness for prevention and early detection amongst the general public, and healthcare practitioners. This may be a challenge in many communities due

to gender issues, religious beliefs, financial burden, cancer misconceptions and more sociocultural factors that may impact how people talk or not about cancer and whether or not they are likely to engage in preventive and early diagnosis behaviors.

History of Cancer

The word Cancer first originated in ancient Greece known as the term Karkinos which translates to crab. The reason why Hippocrates named the disease after a crab is because of the “finger -like spreading projections” of cancer that seemed like that of crab (“The History of Skin Cancer,” 2015). The origin of the term cancer plays a big part on how it is viewed and treated by health care professionals. Around the 17th century people viewed cancer negatively and sometimes it was viewed even as the “wolf” because of how it eats up the human (Matson, 2022). Due to this perception, many health care practitioners treat it violently currently. There has been common war and battle language used around how to fight cancer particularly amongst American Media. This can sometimes cause patients fear and not allow patients to express their conditions to health care providers which can be very damaging to the patients health.

Overview of Societal Perception of Cancer

The negative perceptions can create a rise in stigmas, myths and misconceptions. One of the common misconceptions that arose in all societies is that cancer is a communicable disease which causes patients to be shunned and isolated by society. During the industrial revolution in Britain, cancer was believed to be contagious mainly because groups of people were likely to get cancer together (Magazine & Gershon, n.d.). In reality, the actual reason was that there was an increase in carcinogens and different types of carcinogens, for example smoke in the atmosphere from the industrial revolution (Magazine & Gershon, n.d.). This example primarily shows the

rise in contagious misconception in Britain but the idea of cancer being contagious can come from different reasons in other continents. This paper will focus on the stigmas, perceptions and misconceptions of cancer across different continents. It will also look into how the perceptions negatively affect the patient, and possible solutions to prevent it. A wide range of geographical origins, including South Asian, Chinese, European, American, and Native American origins will be examined in depth. Eastern cultures tend to have conservative outlooks whereas western cultures have liberal attitudes. The contrasting perspectives play an interesting role in how it influences patient behaviors.

Table 1. Most Common Cancers Across Regions

	Male	Female
South Asia	1. Lung	1. Breast
	2. Lymphoma	2. Cervical
	3. Prostate	3. Lymphoma
China	1. Lung	1. Breast
	2. Stomach	2. Lung
	3. Colorectal	3. Colorectal
Middle East	1. Lung	1. Breast
	2. Prostate	2. Cervical
	3. Colorectal	3. Stomach
Europe	1. Prostate	1. Breast
	2. Lung	2. Colorectum
	3. Colorectum	3. Lung
America	1. Prostate	1. Breast
	2. Lung	2. Lung
	3. Colorectal	3. Colorectal

Cancer Perception in South Asia

Introduction

South Asian cultures include Afghanistan, Bangladesh, Bhutan, India, Iran, Maldives, Nepal, Pakistan, and Sri Lanka. Within South Asia the top three cancers amongst male are lung, lymphoma and prostate cancers. Amongst females the top three are Breast, Cervical and lymphoma cancer (Table 1) (Winter et al., 1999). In recent years, there has been a rise in cancer amongst the south asian population due to the increase in tobacco use, air pollution, unhealthy lifestyle, genetics, infections and lack of cancer screening. The primary emphasis will be on lack of cancer diagnosis since the negative perceptions of cancer arise from minimal knowledge and awareness of the disease. Considering the demographic landscape and the unique socio-cultural context of South Asia, countries which will be honed upon are India, Pakistan and Bangladesh.

Table 2. Most Prevalent Types of Cancer Between Male and Female and Crude Rates of Cancer Across South Asia

	Male	Female	Incidence Rates per 100,000
India	1. Oral	1. Breast	96.4 per 100,00
	2. Lung	2. Cervix	
	3. Stomach	3. Ovary	
Pakistan	1. Oral	1. Breast	107.6 per 100,000
	2. Lung	2. Oral	
	3. Esophagus	3. Cervix	
Bangladesh	1. Esophagus	1. Breast	105.5 per 100,000
	2. Lung	2. Cervix	
	3. Oral	3. Esophagus	

India

In India, it is estimated that about 2.7 million people are diagnosed with cancer which means 1 in 9 people are affected by this condition. The crude rate for cancer incidence rate in India is 96.4 people per 100,000 (“Global Cancer Data by Country | World Cancer Research Fund International,” n.d.). The top three cancers between male are oral, lung and stomach cancer and between females they are breast, cervix and ovary (“Cancer Statistics,” n.d.). Statistics show how cancer is an emergent health concern in India. Even with prevention, cultural beliefs and lack of awareness tend to play a big part in the Indian community, which is generally reluctant to engage in preventive and early diagnosis behaviors. Primarily women, rural population, low income population, children and patients with specific cancer are affected by the misconceptions of cancer. A qualitative study was performed in Karnataka where participants with Cervical

cancer and Breast Cancer were interviewed (Nyblade, Stockton, Travasso, & Krishnan, 2017). Researchers reported three common themes from the interviews: 1. Fear of cancer transmission; 2. Blame for causing cancer; 3. Fear of disability from cancer. These themes are what prevents women from getting checked up in the hospital and not receiving early diagnosis. These themes are what causes social stigmas and influences the society to isolate and verbally harass the patient. Some patients even suffer from loss of employment if the company finds out about the health condition (Nyblade, Stockton, Travasso, & Krishnan, 2017). The cancer mortality rate of India in 2020 was 850,000 ("Cancer Statistics," n.d.). One of the biggest reasons why breast cancer mortality rate in India is high is due to the late diagnosis of the cancer. The media also played a huge part in influencing society's perception of cancer by showing how patients with cancer always die on television. Rural societies also attribute cancer to evil spirits or as karma (Kumar et al., 2019). These factors cause patients to lose hope and not take proper treatment anyways because they believe they will die either way. Women with breast cancer also face reduced marriage prospects because society believes that there is a defect with the girl (Kaur et al., 2023). Another barrier causing women to not get checked up is the embarrassment of getting a breast examination performed by a male doctor (Kumar et al., 2019). In India, women are still not able to get enough freedom especially in the rural communities which is the main reason proper early diagnosis and treatment are inaccessible for them. For example in rural India husbands decided the treatment methods in majority of the cases. Due to their disadvantage in Indian society, women frequently lack access to basic healthcare services, leaving them with little alternative but to rely on their husbands for the treatment they need for their illnesses (Singh & Arora, 2008). On the other hand, regardless of gender, cancer patients still suffer from similar myths. These myths also negatively impact one's lifestyle and prevent them from

financially providing for their medical treatment. For example, in rural Assam, India, cancer patients face negative attitudes and actions that result in challenges in their occupational opportunities, the perception of cancer as incurable, the belief that cancer is contagious, and negative behavior such as anger and isolation from the general public towards those with cancer. Due to the fear of these reactions the rural population also tends to delay seeking medical attention (Pak et al., 2021). All of these misconceptions, stigmas and myths highlight that there is a systemic issue with the lack of cancer knowledge and accessibility to health care. In order to resolve these issues, awareness must be increased by involving community leaders from rural villages and addressing cultural beliefs towards cancer hence reducing stigma. Improving access to medical attention is also a significant aspect of removing the barrier. Involving more female health care practitioners for female patients so they aren't embarrassed of seeking medical care.

Pakistan

Around 178,388 new cases of cancer emerged in Pakistan in 2020 and the number is still on the rise. Around 107.6 people per 100,000 people are affected by cancer (“Global Cancer Data by Country | World Cancer Research Fund International,” n.d.). Amongst women, the top three prevalent cancers are breast, oral and cervix and amongst men the top three common cancers are oral, lung and esophagus (“Cancer Today,” n.d.). Pakistan has similar trends with India regarding the commonalities of the cancer types. These may occur due to similar perceptions towards women and taboo regarding women’s health condition. The three major themes regarding breast cancer diagnosis and treatments amongst women are sociocultural, individual and structural barriers (Saeed, Asim, & Sohail, 2021). Individual barriers have been raised from the lack of awareness, social support and spiritual healing. Religious beliefs give rise to religious misconceptions, and social pressure which have influenced the diagnosis of cancer

stage due to late screening. Individuals believed spiritual spells such as Dam-Darood would aid in the healing of cancer in their body (Saeed, Asim, & Sohail, 2021). Socio-cultural barriers preventing them from screening are feminine sensitivity, stigmatization and aversion to male doctors (Saeed, Asim, & Sohail, 2021). The structural barriers include lack of financial resources and apathetic medical services (Saeed, Asim, & Sohail, 2021). Hospitals in Pakistan are lacking cancer screening facilities, oncologists and basic lack of medical infrastructure for cancer patients. Along with India, Pakistan also shows a correlation between education level and knowledge of cancer. Amongst the educated and literate population, information related to cancer is accessible, however the low income families struggle with gathering valid information (Kumar, Shaikh, Khalid, & Masood, 2010). They often tend to believe in myths and taboos related to cancer. Easy access to spiritual leaders, or faith healers may give rise to the myths and stigmas which ultimately can deteriorate the patients health and prevent from diagnosis. Females were more affected by the myths especially relating to cancer being caused due to god's curse (Kumar, Shaikh, Khalid, & Masood, 2010). India and Pakistan have similar themes of why patients struggle to get proper medical care and their misconceptions tend to rise from the lack of knowledge and accessibility to cancer resources.

Bangladesh

Around 156,775 new cancer cases arose in Bangladesh in 2020. In Bangladesh 105.5 people per 100,000 are affected by cancer ("Global Cancer Data by Country | World Cancer Research Fund International," n.d.). A lot of the time due to lack of financial ability, awareness, and freedom may cause patients to unknowingly die from cancer, which is why the incidence rates may have discrepancies. Amongst the male population, esophagus, lung and oral were the most prevalent cancer whereas breast, cervix and esophagus is common in females ("Cancer

Today,” n.d.). The commonality of breast and cervical cancer in all three countries (India, Pakistan and Bangladesh) essentially shows the lack of accessibility for women to get proper medical treatment. Cervical cancer, which is caused by HPV viruses, is easily preventable with proper screening (e.g. Pap smears) and vaccination. The common themes why females have barriers to breast cancer screening are inadequate knowledge of breast cancer, poverty, inadequate medical care, counter productive beliefs, social norms limiting women’s bodies, and neglect by family. The counterproductive beliefs which delay women from getting checked up include the fear of being diagnosed with cancer, the perception of cancer as synonymous with death, the misconception of cancer as contagious leading to isolation, restrictions on personal freedom, and the societal taboo surrounding breast cancer. These factors prevent women from vocalizing their issues, hence stopping them from being diagnosed early for cancer. The delay in diagnosis ultimately is what causes cancer to be a death sentence which continues the cycle of defining cancer as incurable. Similar to India and Pakistan, lung and oral cancer is prevalent amongst male. A lot of the reason why lung cancer and oral cancer is caused by tobacco or smokeless tobacco. In South Asia as a whole, smokeless tobacco is very common amongst the general population due to the easy accessibility of the product, its addictiveness, peer pressure, and misconceptions of beneficial use of smokeless tobacco (Kakde, Bhopal, & Jones, 2012).

Overview and Possible Resolution to Cancer Stigmas in South Asia

The literature review analysis shows how the lack of information of the cancer itself gives rise to misconceptions which perpetuates stigmas in the community and prevents patients from getting proper medical care ultimately causing certain cancer type cases to increase. These misconceptions mostly arise from the cultural beliefs, lack of medical infrastructure, financial burden, and stigmatization. One significant commonality among the three countries is that

gender discrimination has been a major contributing factor to cancer-related deaths, primarily due to limited access to diagnosis and treatment. To help reduce stigmas within the community, people in South Asia have an opportunity to focus on creating education and awareness conferences in rural areas, screening programs, improving medical infrastructure, and also addressing and debunking smokeless tobacco myths.

Table 3. Visual Representation of Cancer Perceptions in South Asia

Common Cancer Themes	Causes	Results	Final outcomes
Lack of awareness and education	1. Diet 2. Unaware of early cancer signs 3. Lack of freedom (accessibility to health facilities) 4. Linguistic Barriers 5. Myths and Misconceptions 6. Lack of trust in healthcare system	1. Delayed diagnosis of cancer 2. Isolation towards cancer patients 3. Verbal abuse towards cancer patients 4. Loss of employment 5. Discrimination 6. Loss of family support 7. Loss of marriage prospects	1. Cancer is not detected when it is still curable 2. Increase in fear, anxiety and depression 3. Increase in cancer in south asian societies
Stigma and cultural beliefs	1. Cancer=Punishment 2. Cancer=contagious 3. Cancer=incurable 4. Cancer=Taboo topic		
Limited medical infrastructure	1. Limited healthcare resources 2. Lack of cancer screening facilities 3. Lack of access to cancer treatment 4. Lack of transportation to infrastructure		
Financial constraints	1. Lack of health insurance 2. High treatment cost 3. Lack of government funding 4. Loss of income 5. Transportation cost		

Cancer Perceptions in China

Introduction

Unsurprisingly, incidence rates and mortality rates are on the rise in China as well. China had 4,568,754 new cancers in 2020 and its numbers of new cases are on the rise. Around 203.8 per 100,000 people were affected by cancer in 2020 (“Global Cancer Data by Country | World Cancer Research Fund International,” n.d.). Amongst women, breast cancer, lung and colorectal cancer are the top three cancers and amongst men, the common cancers are lung, stomach and colorectal cancers (“Cancer Today,” n.d.). Factors which cause the upcoming cancer cases are lifestyle, dietary habits and air pollution. Even though China is an advanced and modern country, a segment of the population still contains cultural factors which continue to stigmatize cancer. Ultimately these negative perceptions cause delayed detection of cancer which contributes to the rise in cancer incidences in China.

History of Cancer in China

Unlike South Asian countries, cancer is more widely discussed due to references made in traditional Chinese medicine literature. Traditional Chinese medicine heavily focuses on strengthening body resistance, eliminating pathogens and ultimately focusing on cancer prevention. Due to these values, traditional Chinese medicine has laid the foundation of current modern Chinese medical infrastructure for cancer (Liu, Wang, Zhang, Fan, & Lin, 2015). Even Though traditional Chinese medicines have been supportive of cancer treatment, “a stoic, conservative culture” still exists in Chinese communities which makes it tough for patients to open up (“Death By Stigma,” n.d.).

Stigma Surrounding Cancer in China

Since there is a huge emphasis of traditional Chinese culture in everyday lifestyle, physical and emotional care, cancer is often seen as a patient's failure. Disease in traditional Chinese culture is also associated with “personal weakness” hence why stigmatization and judgment between Chinese societies arises regarding cancer (Lui, Ip, & Chui, 2009). A lot of the stigma arises from the patients themselves (Liu et al., 2016). According to a study focused on lung cancer patients coping with stigmas, reasons why stigmatization occurred were due to understanding cancer as a consequence of smoking, loss of employment from companies, loss of ability to help support family and damage to self image. Most of the stigmatization comes from the patients themselves and they tend to blame themselves which eventually causes a rise in depression amongst patients. Society also proceeds to take negative actions such as discrimination, and isolation which causes patients to not talk about their sickness freely. Similar to South Asian countries, cultural beliefs exist which prevent cancer patients from disclosing their symptoms to the doctors or families. Breast cancer patients face stigmatization from Chinese societies due to the belief that it is caused by bad Feng Shui or karma. Similar to lung cancer patients, breast cancer patients also suffer from poor body image. These stigmas affect patients' self efficacy and demotivate them from getting better (Jin, Xie, Zhang, Gong, & Zhang, 2021).

Overview of Cancer in China and Possible Resolutions

Even though speaking up about cancer is easier in China than in other conservative countries, Chinese societies also stigmatize cancer and negatively impact patients mental wellbeing and self image. On the bright side, traditional Chinese medicine has always been

supportive of cancer treatment plans which is an advantage for patients to disclose their issues and get diagnosed soon, however certain communities exist which perpetuate cultural misconceptions. Other than society's negative perception of cancer, a big chunk of stigmatization comes from self negativity, which further propels that mindset. Body image problems are a huge chunk of where self stigmatization comes from. A possible reason why this may be the case is because Chinese communities “are strongly focused on appearance” (“What’s Behind China’s Beauty Addiction?,” 2019). Possible resolutions to reduce these stigmas are by holding awareness campaigns, and support groups with other cancer patients and cancer survivors to help cancer patients relate to others and improve self efficacy.

Table 4. Common Perceptions of Cancer in China

Negative perceptions	Results in	outcomes
Personal weakness, failure	Experience of guilt, shame, self blame, isolation, discrimination, increased stress and anxiety	Delayed cancer diagnosis, depression, anxiety, low self esteem
Self-condemnation	Experience of guilt, shame, self blame	
Societal stigmas	Isolation, loss of employment	
Misconceptions from cultural beliefs	Reduced social support, isolation	
Body image	Reduced self esteem, self affection, self isolation	

Cancer Perception in The Middle East

Introduction

The Middle East also has significant amounts of death due to cancer annually. In the Eastern Mediterranean region, Cancer is one of the top 4 leading causes of death. Amongst women in the Arab world almost 244,317 new cancer cases rose (Mahdi et al., 2022) and amongst men 217,203 new cancer cases emerged (Mula-Hussain et al., 2022). Around 116.2 per 100,000 women are affected around 2020 (Mahdi et al., 2022) and 133.5 per 100,000 amongst males (Mula-Hussain et al., 2022). The leading cancer types in women and men respectively were breast and lung cancer. The other two leading cancers among men were prostate and colorectal and women were diagnosed with cervical and stomach (Arafa, Rabah, & Farhat, 2020). Similar to South Asia, the Middle East also falls under the conservative countries which implies strong cultural or religious beliefs. The religious and cultural beliefs have negative and positive impacts on cancer which will be discussed.

Cultural, Religious and Social Factors Contributing to Cancer Stigma in The Middle East

Most cultural factors previously discussed were overall negative, however there are positive effects to some religious/cultural beliefs. On a larger scale Arab countries have lower cancer rates compared to other parts of the world. Religious traditions such as fasting, and forbidden activities like smoking and alcohol play a huge role in the low mortality rate of cancer compared to other countries (Arafa & Farhat, 2022). Fasting is beneficial to treating and preventing cancer because it can slow cancer growth and promote cell regeneration via lowering glucose levels which starves cancer cells. During the process of chemotherapy, cells which are primarily targeted are the ones that grow and divide quickly. Normal cells do not proliferate

during fasting whereas cancerous cells still proliferate which is why chemotherapeutic medications are able to target cancer cells quickly. Alcohol is forbidden in Islam and cigarette smoking is lower in the Middle East which is also why cancer incidence rates are lower. A study on tobacco use in the Arab population has revealed that the prevalence of tobacco use is highest among adolescent boys between the ages of 13 and 15, with between 11% and 34% of them having traces of tobacco in their system. Among male youth in Arab communities, the consumption of tobacco and alcohol is frequently attributed to peer pressure (Obermeyer, 2015). According to a recent study, the prevalence of smoking among men is 34.7% higher than that among women (Sibai et al., 2016). Given the prevalence of smoking among men in the MENA region as shown by numerous studies, it appears that this trend may contribute to the high rates of lung cancer among men in the region.

Middle Eastern Societies tend to stigmatize the patients and cause a barrier for patients to access proper cancer facilities. The common misconceptions that arose when women were being interviewed were that cancer is a test or is a form of punishment from God (Salem & Daher-Nashif, 2020). In order to cope, women would pray more often. Religion can play a positive role. A recent study suggests that individuals who demonstrate higher levels of religiosity and spirituality may experience a reduced risk of developing depressive symptoms compared to those with lower levels of religiosity and spirituality. (Dewan, Lyons, Song, & Hassounah, 2022). Another barrier is that women may not be interested in getting screened by male doctors (sometimes even female doctors) since modesty is a virtue in Islam (Donnelly et al., 2013). Another study showed how Arab men view breast cancer and how they behave to families who are affected by breast cancer. Arab men are supportive and understand the importance of early screening, however they do want the health care provider to be female

(Donnelly et al., 2017). Another barrier for women is that they have to seek men's permission before going to hospital ("CHALLENGING INEQUALITY," n.d.). This also causes delays in diagnosis. Stigmas in the middle east region mostly affect women because the middle east region is still mostly patriarchal. Similar to the South Asian countries, the Middle East also viewed cancer as a death sentence Another notable difference is that in the middle east often times the cancer patient is not informed of the condition of their health or cancer progress in their body in order to protect the patient's psychological well being which is known as the truth disclosure policy (Khalil, 2013).

Possible Resolutions

Compared to the other societies, the Middle East showed religious benefits regarding cancer treatment and prevention. On the other hand the negative perception on cancer mostly affected the women in comparison to men. The main issue women face is the lack of freedom to go to medical facilities. Studies have shown that many patients are utilizing coping mechanisms, such as prayer, to deal with the challenges of their medical conditions, and that these coping strategies have had a positive impact on their emotional well-being. The policy of truth disclosure not only impacts the lifestyle of patients but also violates medical ethics, potentially affecting cancer progression in the patient. Possible resolutions for these barriers and stigmas are holding public awareness campaigns with religious representatives, more female physicians, family counseling services so that the whole family gains cancer knowledge and offer support to the patient.

Table 5. Visual Representation of Cancer Stigmas in The Middle East

Themes	results/ coping mechanism	
cancer=punishment	Consistently praying	May alleviate depression
Modesty	Lack of female healthcare staff→ barrier	Delayed diagnosis
Conserve a truth disclosure policy	intended purpose: patients wellbeing	Possible harm if lifestyle changes→ increase cancer progression Medical ethics violation
Lack of freedom	Inaccessibility to medical facilities	Delayed diagnosis or no diagnosis at all

Cancer Perception in The Western World

Introduction

The western world mostly consists of the USA, and Europe. These countries were grouped together mainly because they hold similar social, cultural and political values. Western societies tend to have more open communication during the screening and treatment process of any cancer.

In 2019, Europe faced 1.9 million cancer deaths. Amongst male patients, the top ranking cancers were prostate, lung and colorectum (“Europe,” n.d.). On the other hand, the top 3 leading cancers amongst women were breast, colorectum and lung. Since there has been an increase in immigrants from across the world in Europe, it is no surprise that some cultures can perpetuate cancer related stigmas (“Cancer Today,” n.d.). The history of cancer in Europe needs to be studied in order to see the cancer perception before there was an rise in immigrants..

The USA had 1.9 million cancer cases in 2022 (Siegel, Miller, Fuchs, & Jemal, 2022). The leading causes of cancer in 2020 were mostly prostate, lung and colorectal cancer amongst

men whereas in women the top leading cancers were breast, lung and colorectal cancer (“Cancer Today,” n.d.). Similar to Europe, the USA is also filled with different cultures which may cause an influence in cultural stigma between the general public. The USA's evolution of cancer perception will also be looked at along with history to understand how cancer misconceptions are being debunked or formed. Along with the general public in the United States, Native American populations and their perceptions of cancer will also be evaluated. Since Native Americans were the first residents of North America, their perception of cancer differed significantly from the Americans who migrated from Europe back in the day which is why it is important to study the cancer perception amongst Native American communities as well.

Historical Perspective on Cancer Perception in Europe

Other than dysentery and bubonic plague, cancer fell under one of the major diseases which affected the medieval population. Due to lack of cancer knowledge, cancer treatment options were very limited during the medieval times. However once the industrial revolution occurred, cancer incidence rates spiked. Even in Europe around the 1500s, the French defined cancer as an incurable disease due to the lack of advanced medical technology (Magazine & Gershon, n.d.). In early modern Europe, the term cancer would also be used in nonmedical context to show the negative connotation of any matter. For example the Nazis used to label inferior classes as cancer, which shows how cancer has always been a negative term to associate with anything undesirable occurring (Stolberg, 2014). Cancer is also perceived as a silent killer to show the unpredictable nature of it. Based on these observations, cancer has always been looked at in a negative context which shows that negativity is ingrained within European culture which may cause stigmatization.

Current Cancer Perception in Europe

Europe is now a continent filled with a diverse population, which means that cultures must have been spread and affect other communities' perception on cancer, which may positively or negatively influence cancer patients. Since smoking is widely known to cause lung cancer, a significant amount of the general population stigmatize lung cancer by believing it as a self-inflicted disease (Berghmans et al., 2020). Patients may have stress and anxiety from the negative perception and ultimately are neglected of psychosocial needs compared to other cancers which may not be affected by various lifestyle factors. Lung cancer patients face high amounts of scrutiny and stigmatization because the communities blame the patients for their own disease which causes them to be depressed and not voice out their condition freely. High incidence rates of lung cancer may be a result of delayed diagnosis, negative mental state affecting the body, or refraining from discussion conditions.

Based on the statistics, breast cancer is also one of the leading cancers in Europe. Surprisingly, breast cancer stigma is lower in Europe compared to the eastern continent. This may be the case due to awareness organizations. These organizations actively participate in advocacy to educate the general public about breast cancer. Currently Breast cancer civil society organizations in Eastern Europe are developing strategies on how to advance breast cancer advocacy and improve cancer care (Antone, Kizub, Gralow, Zujewski, & Dvaladze, 2021). They focus on removing stigma relating to fear of disease, cultural taboos and myths and providing more campaign programs to increase cancer knowledge (Errico & Rowden, 2006). While cancer advocacy programs do a good job in making a safe space for talking about cancer, men with breast cancer face backlash in European societies. Men feel excluded, discrimination and rejected to visit breast cancer programs. Some men also go through self stigmatization because

they feel that they are worthless for their families. This can cause men to conceal their symptoms regarding breast cancer which will lead to delayed diagnosis (Midding et al., 2018). Since Europe and the USA both have progressive attitudes compared to the rest of the countries mentioned above, they have similar knowledge towards cancer. A contrasting factor between the two regions is that the USA has a history of Native American occupation, which may have influenced the cancer perceptions within that community.

Native American Perception of Cancer (Pre-Columbian Era)

During the Native American Columbian Exchange and Pre Columbian era, Smallpox, Measles, Whooping Cough, Chicken Pox, Bubonic Plague, Typhus and Malaria were the diseases which were spread (Nunn & Qian, 2010). There is currently not much cancer data on the web during the pre columbian or the Native American columbian exchange times. However a lot of the trauma natives have from colonizers still exist to this day (Cordova-Marks, Carson, Monetathchi, Little, & Erdrich, 2022). When they hold negative perceptions towards the healthcare system, this can lead to a lack of trust in the system. As a result, they may be less likely to seek timely diagnosis and receive proper treatment, ultimately impacting their overall health outcomes. (Guadagnolo et al., 2009). There are different categories of stigma which exist among the native american communities which are public stigma, structural stigma towards “self induced” cancer and self induced stigma (Else-Quest & Jackson, 2014). Public stigma causes a barrier via isolation where patients are excluded and avoided by friends and social circles. Structural stigma occurs due to lack of financial help from the government towards “self induced cancer” which causes a delay in diagnosis and self blame and mental health. According to some communities lung cancer is a self induced cancer due to the patient's past smoking history.

Native American men tended to have liver, stomach, kidney, colorectal, lung cancer, myeloma whereas native American women have liver, kidney, colorectal and cervical cancer (CDC, 2022). Amongst women there are fears of social judgment, rejection, self blame and shame which stigmatizes cervical cancer (Peterson et al., 2022). Although both men and women can be affected by cervical cancer, as it is transmitted through sexual contact, women often face greater societal judgment in relation to the disease. Patients often taking the vaccine or seeking cervical cancer treatment are stigmatized based on the perception that cervical cancer is occurred to those with more sexual partners. This leads to society shaming and blaming women which causes women from refraining their symptoms or even getting detection tests just as a regular check as recommended by health care providers.

Current Cancer Perception in America

Cancer is the second leading cause of death in the USA. The best way to decrease cancer incidence rates is by focusing on prevention and diagnosis methods (“American Cancer Society Sets Goal to Reduce US Cancer Death Rates 40% by 2035,” n.d.). Around 297.3 per 100,000 people are affected in the US by cancer (“Global Cancer Data by Country | World Cancer Research Fund International,” n.d.). A significant way in which prevention of cancer can be accomplished is by spreading cancer knowledge amongst the community which allows them to seek medical attention and participate in early cancer detection events. Amongst women the common types of cancer are breast, lung and colorectum whereas amongst men its prostate, lung, and colorectum (“Cancer Today,” n.d.). Since the USA is not a conservative, there aren't prevalent stigmas preventing patients from seeking medical help. However the diverse nature of the country may have microstigma which may internally affect patients from expressing their concerns.

The USA may have advanced technology, modern ideas and liberal perception, but multiple ethnic communities exist within it which may have conservative viewpoints, societal pressure and traditional customs. These behaviors and actions influenced by different cultures can stigmatize certain cancers which can prohibit patients from accessing healthcare services. For example Chinese American breast cancer patients who aren't as much assimilated to the american culture are more likely to face psychological distress and self stigma (Warmoth, Wong, Chen, Ivy, & Lu, 2020). The common emotional response to cancer between the ethnic communities are fear, underestimation (taking the disease lightly), fatalism, and pessimism and common negative perceptions of cancer are believing that cancer is incurable, and cancer was created by god (Dein, 2004). There is also a direct relationship between cultural beliefs and low usage of mammography which would cause patients to not be diagnosed with cancer sooner (Dein, 2004). A significant portion of the US population are immigrants from different ethnic backgrounds which explains the diverse cultural set and their negative effects on cancer perception. The media also plays a huge role in stigmatization of cancer, however the stigma in this case can affect the patient positively or negatively. The media puts a lot of emphasis on how to combat cancer. They use warlike metaphors to encourage cancer patients. Cancer is also used as a metaphor to describe negative circumstances unrelated to medicine (Soffer, 2022). The use of military metaphors for cancer especially is harmful and stigmatizing because it raises public fear which can prevent patients from getting proper diagnosis. If patients are unable to “battle” cancer they may tend to self blame which can lead to mental health problems. The use of battle-related metaphors in the context of illness may encourage healthcare professionals and patients to pursue aggressive treatments that may not necessarily be in the best interest of the patient. A lot of the stigmas in the US may be structural also which focuses more on the lack of

health care facilities for the marginalized communities. The black population tends to die more from breast cancer than the white population due to the lack of access to healthcare (Yedjou et al., 2019). For example black women tend to be twice as more uninsured in comparison to white women.

Possible Solutions to Combat Cancer Stigma in The Western World

Western World is essentially a big melting pot for different cultures which is a great avenue for different cultures, traditions and customs to spread. Different cultures already have existing societal viewpoints of cancer. Most perceptions of cancer from ethnic communities pose a limitation towards patients to freely seek medical help. In order to reduce the stigma, substantial changes in the healthcare facilities, and more awareness programs need to be made. A key purpose of awareness programs is to debunk the myths associated with cancer. These programs ultimately can reduce stigma and prevent isolation from family and society. Another solution to tackle the negative cancer metaphors used by the media and general public is by using empowering and patient centered language, educating healthcare staff, using person-first language (“person with cancer” rather than cancer patient”), and being mindful of cancer metaphors like battling cancer or fighting cancer. There aren't alternative helpful metaphors however phrases which empowers and respects the autonomy of patients with cancer is a better way to encourage the patient. Applying these solutions in the long run may lower cancer mortality rates and may prevent mental health problems.

Table 6: Perception of Cancer in The Western World

Region	Common Themes	Result
Europe	Self inflicted disease Fear of disease Cultural taboos discrimination	Psychological distress which causes patients to not voice out their condition
Native Americans	Mistrust of health care system Self inflicted disease Social judgment	
America	Self stigma Ethnic cultural taboos Media cancer metaphors	

Conclusion

Cancer is one of the top five leading causes of death across the world. Several factors affect cancer such as environment, diet, lifestyle, genetics, age, and infections. An underrated factor which is not talked about much is the sociocultural aspect of cancer. The sociocultural aspect of cancer focuses on cancer perceptions across different cultures such as myths and misconceptions, which may sometimes stigmatize cancer. The sociocultural factor impacts the cancer incidence significantly because the stigmatization of cancer can stop patients from accessing medical help, which can cause late diagnosis of the cancer status causing the cancer to be metastatic. The stigmatization of cancer caused self blame, isolation, body image issues, and vulnerability. These results are caused by misconceptions such as cancer being a self-induced disease, death sentence, contagious, or punishment of bad behavior. The misconceptions affect the societal stigma of cancer which ultimately delays patients' cancer diagnosis. Future studies can delve more deep into detailed methods to reduce cancer stigma and advocate for prevention and diagnosis of cancer.

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